

TP: Photo-Electron Spectra measurement of Krypton Gas

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INTRODUCTION

In this study, we delve into the electronic intricacies of Krypton gas using Photoelectron Spectroscopy (PES), a powerful technique that unveils the fundamental properties of matter. Employing High-Harmonic Generation (HHG), our objective is to unravel the spin-orbit splitting of Krypton electrons, providing crucial insights into the behavior of these atoms under intense laser fields. The challenges lie in the ultrafast timescales and precision required to capture attosecond dynamics, revealing the inherent complexities of photoionization processes. This study positions itself within the state-of-the-art by recovering kinetic energy peaks through Gaussian fitting of the experimental data and comparing it to the precedent literature.

METHODS

TO obtain a spectrum of the Kinetic energies of photoelectrons we have used the setup illustrated in the figure below. An original pump laser of Titanium-Sapphire (TiSa) laser emitting near-infrared (NIR) pulses at a wavelength of $\lambda_i = 800\text{nm}$ was used. The TiSa laser operated with a power of 2 millijoules (mJ) and a frequency of 5 kilohertz (kHz). Our apparatus is composed of two isolated main chambers: Higher-Harmonic-Generation (HHG) chamber where Xenon gas is introduced in a controlled pressure environment and a spectrometer encased in a chamber where the Krypton gas is led.

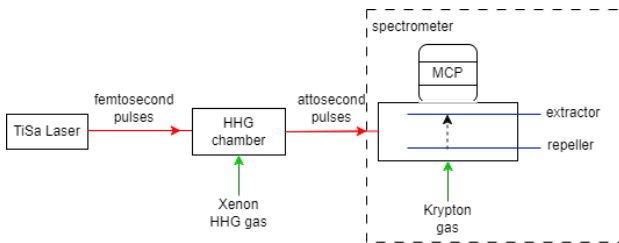


Fig. 1. diagram of the PES setup

A. HHG using TiSa NIR Pulses

Within the HHG chamber, the original TiSa laser beam interacts with the Xenon gas. The strong laser field leads to the ionization of Xenon atoms and subsequent high-harmonic generation, producing harmonics in the extreme ultraviolet (XUV) region much like in the experiment of V. Loriot *et al* [5]. How does this phenomenon takes place? At rest, an

electron within the Xenon atom doesn't have enough energy to lift itself out of the potential well that atom's nucleus is generating. However, the NIR Laser's E -field distorts the atom's potential field, lowering the barrier and electrons may escape there bounding potential via quantum tunnelling [4]. It has acquired more energy than necessary for it's escape so it has excess energy in form of kinetic energy. Since the laser field oscillates it changes direction and the newly free-electron is pulled back in the direction of it's escape. Upon returning to the parent nucleus it can then emit bremsstrahlung-like radiation called *overtones* in the XUV domaine during a recombination process with the atom as it returns to its ground state. This description has become known as the recollisional model of HHG. A scheme of the model is represented in Figure 2 [1].

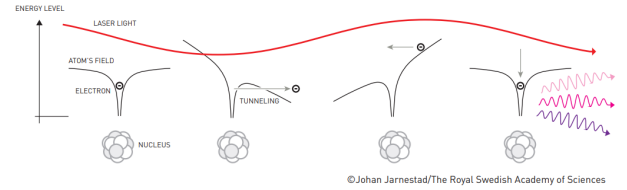


Fig. 2. Recollisional model of HHG

The Laser emits in the Near-infrared, femtosecond pulses, however, in order to probe the fine structure of the sublevel electron shells we need get to the attosecond time domain. Xenon atoms are used in the previously described HHG process. At the output of the HHG chamber we have a XUV beam with attosecond pulses and residual NIR of the unprocessed beam. We use an Aluminium plate to absorb the NIR.

B. The spectrometer

The second sealed chamber has a charge accelerator and a gas inlet. The inside of the chamber is in a controlled near-vacuum state at 10^{-6} mbar, in order to minimize interaction with order materials. An extractor plate is fed with $V_{ex} = -2900\text{volts}$ and a repeller plate with $V_{rep} = -2200\text{volts}$. They are both placed facing each other, parallel to the incoming beam. The laser ionises the Krypton gas, which emits electrons with different Kinetic energies depending on their energy states. They are accelerated then in the direction perpendicular to the beam's, in direction of the extractor and received by a Microchannel plate(MCP) mounted with

a CCD camera adjacent to the sealed chamber. Depending of the Kinetic energies of expelled electrons via the photoelectric effect, the distribution of the electrons varies, and their according levels of energies can be observed on the CCD.

C. Gaussian fitting

We proceed to fit the peaks obtained with a Gaussian profile in order to get the parameters such as the maximum amplitude, position of the peak(mean), sigma and deduce the FWHM and Resolution of the spectrometer. By hand picking the right inflection points of the curve we can deduce the lower and upper bounds to be used in the fitting process. In the figure 3 we have used [16-16.8] eV to fit the first peak.

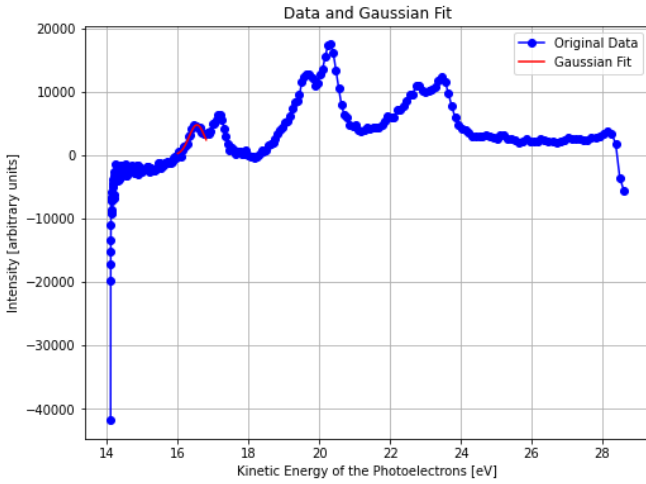


Fig. 3. Fitting each peak with a Gaussian

The code that we used fitted the curve using a custom function defined with the following equation:

$$G(E) = A \cdot \exp\left(-\frac{(E - E_0)^2}{2 \cdot \sigma^2}\right) \quad (1)$$

The equation (1) refers to a simple gaussian profil with A , E_0 , σ as the parameters that have been found using the `curve_fit` method applied to the custom gaussian method in the `scipy.optimize` library on python [6]. A refers to the maximum amplitude at E_0 the position of the peak, and σ the standard deviation. We evaluated the Full Width at Half Maxima with (2) and the Resolution R of the corresponding peak with (3).

$$\text{FWHM} = 2\sqrt{2 \log(2)} \cdot \sigma \quad (2)$$

$$R_{\text{peak}} = \frac{\text{FWHM}_{\text{peak}}}{E_{0,\text{peak}}} \quad (3)$$

The plots in the Results section were included with the Confidence intervals computed using the equations (5), (6) which yields the lower and upper bounds of the interval. Additionally the standard Error(SE) in equation (4), is calculated for using (5), (6). We have chosen the Z-score as 1.96, Number of data points as 5 for our calculations, since we have 5 data sets.

$$\text{SE} = \frac{\text{Standard Deviation of the Data}}{\sqrt{\text{Number of Data Points}}} \quad (4)$$

$$\text{Lower Bound} = \text{Mean} - (1.96 \times \text{SE}) \quad (5)$$

$$\text{Upper Bound} = \text{Mean} + (1.96 \times \text{SE}) \quad (6)$$

D. Spin-orbit splitting

Spin-orbit coupling is a relativistic interaction that arises from the coupling of an electron's spin angular momentum with its orbital angular momentum. In the context of PES, which measures the electronic structure of atoms in gaseous state, the spin-orbit coupling can play a significant role. It leads to the splitting of degenerated electronic levels. This splitting is known as the fine structure splitting and has been previously studied on Krypton by Stefan Pabst *et al* [7]. It results in multiple closely spaced peaks in the photoelectron spectrum. Each peak corresponds to a different electron sublevel with a specific combination of total angular momentum J , which includes both orbital and spin angular momentum.

Before considering the fine structure let us consider the outer-most electronic shells of the Krypton atom. Since they are the most weakly-bound ones to the nucleus, they are the first ones to be extracted. Taking the electronic structure of the Krypton : $[\text{Ar}] 4s^2 3d^{10} 4p^6$, into consideration, we can identify the first extracted electrons as originating from the 4p orbital. The p sub-shell can accommodate up to three orbitals, and these orbitals have the same energy when they have the same Principal Quantum Number n and the same Azimuthal Quantum Number l , and are in the same sub-shell p.

The degeneracy arises because the three p orbitals within the same subshell (n and l values are the same) have the same energy. This degeneracy is a consequence of the spherical symmetry of the electron cloud around the nucleus. In the absence of external influences, all three p orbitals in a given subshell have the same energy and are considered degenerate. The Magnetic Quantum Number m_l : Specifies the orientation of the orbital in space. For p orbitals, m_l can take values of -1, 0, and 1. These are summarised in the table I.

Orbital	m_l	m_s	Notation
$4p_{-1}$	-1	$+\frac{1}{2}$	$4p_x$
$4p_0$	0	$+\frac{1}{2}$	$4p_y$
$4p_1$	1	$+\frac{1}{2}$	$4p_z$

TABLE I
QUANTUM NUMBERS FOR THE 4P SUBSHELL

When we consider the spin-orbit coupling, the $4p_{1/2}$ and $4p_{3/2}$ sub-shells can be identified, considering J , and we will proceed to undertake the the measurement of ΔE_{SO} the difference of energy between these two sublevels of energy. A summary with notation is given in table II. As we have generated HHG to probe the sample, the spectra response of the photoelectron emission is concordant. Thus within the spectra the same sub-level originated electrons will appear multiple times. In a previous experiment, we have identified the spectra Harmonic peaks as 11, 13 and 15.

Subshell	Total Angular Momentum (J)	Spectroscopic Term
$4p_{1/2}$	$J = 1/2$	$^2P_{1/2}$
$4p_{3/2}$	$J = 3/2$	$^2P_{3/2}$

TABLE II
SPECTROSCOPIC TERMS FOR THE $4p_{1/2}$ AND $4p_{3/2}$ STATES

For the uncertainty of the spin-orbit splitting energies presented in the table III we have used the equation(7) expressed in [3]:

$$\sigma_{\text{splitting}} = \sqrt{\sigma_{\text{Peak } i}^2 + \sigma_{\text{Peak } i+1}^2} \quad (7)$$

The $\sigma_{\text{ith Peak}}$ is the standard deviation of the i 'th peak, for the Average values of the i^{th} column within the table III.

RESULTS

We have collected various intensity versus photo-electron kinetic energies for the Krypton gas, as visible on the figure (4).

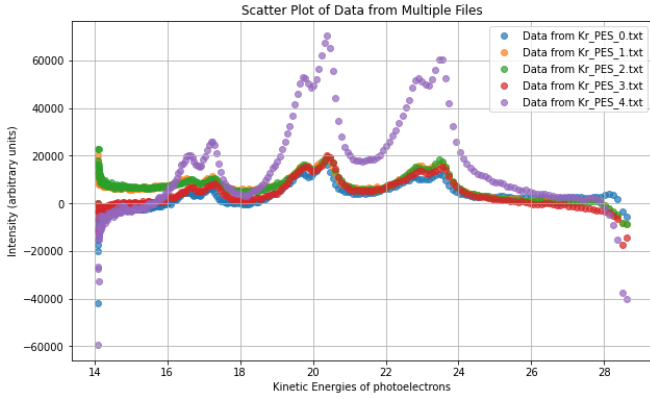


Fig. 4. Scatter Plot of multiple PES data points

E. Kinetic Energies and the S.O. Splitting results

By fitting the 5 datasets of the Krypton PES with the Gaussian fit discussed in the previous section, we have extracted a series of Amplitude, electron Kinetic Energy, standard error to mean (directly provided by and inherent to the fitting method), standard deviation, FWHM and Resolution of the measurement. 4 of the 5 fitting results are provided in the Annex.

The table III recapitulates the means recollected, the average done over the datasets, and the Splitting Energy with uncertainties.

Peaks Means [eV]	1	2	3	4	5	6
Kr_PES_0	16.54	17.14	19.73	20.26	22.81	23.38
Kr_PES_1	16.51	17.18	19.91	20.35	22.97	23.45
Kr_PES_2	16.63	17.19	19.82	20.35	22.91	23.45
Kr_PES_3	16.67	17.21	19.84	20.37	22.94	23.46
Kr_PES_4	16.61	17.18	19.87	20.33	22.92	23.44
Average	16.592	17.18	19.834	20.332	22.91	23.436
Std_deviation	0.0588	0.0228	0.0602	0.0382	0.0540	0.0287
Splitting Energy (eV)	0.588 ± 0.0630		0.498 ± 0.0712		0.526 ± 0.0611	

TABLE III
PEAK POSITIONS AND SPIN-ORBIT SPLITTING ENERGIES OBTAINED BY GAUSSIAN FIT

The figure 5 refers to the table III's average row, where those are taken as the main peak positions and plotted with their corresponding confidence intervals.

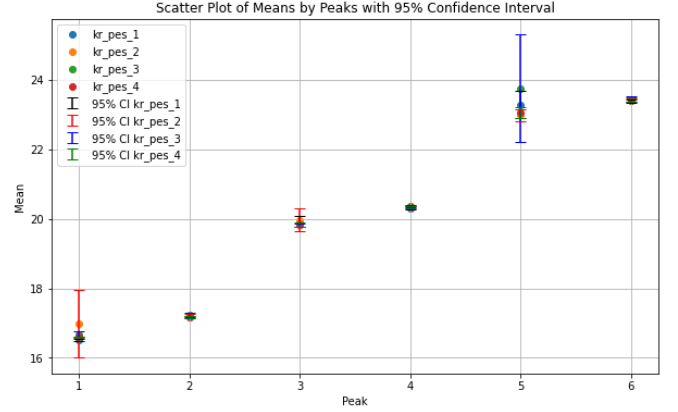


Fig. 5. Plot of E_0 (Kinetic Energies) with their 95 percent Confidence interval vs their peak number

F. Resolutions

We lastly plotted the Resolution per peak number, and added the 95 percent confidence interval computed with the method discussed earlier as the error bars, as seen in figure 6.

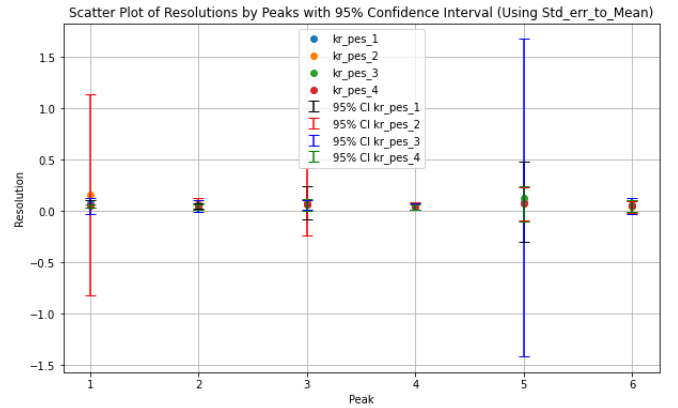


Fig. 6. Resolution by peak including the 95 percent Confidence Interval

DISCUSSION

We have measured previously the SO splitting energy (ΔE_{SO}) over 3 different harmonics corresponding to the energy between $4p_{1/2}$ and $4p_{3/2}$: $\Delta E_{SO,H11} = 0.588 \pm 0.0630$, $\Delta E_{SO,H13} = 0.498 \pm 0.0712$, $\Delta E_{SO,H15} = 0.526 \pm 0.0611$. However how can we be sure of the validity of those values? We need to consider the deduced resolutions in the figure 6, and the literature of the previous experiments. Another thing to notice is that the uncertainties are inherent to the data and not the physical measurement system, so the uncertainties presented are in reality far more superior, but the complexity of the system and the small time-frame where we in touch with the experiment, retained us to develop this aspect further.

For a good understanding of the relation between energies involved in the processes that we observe, one can appreciate

the figure 7, where the ΔE_{SO} within the electron cloud is accessed by a simple difference between the splitted peaks one after the other. The red arrows are the measured Kinetic energies, whereas the blue ones are B.E.

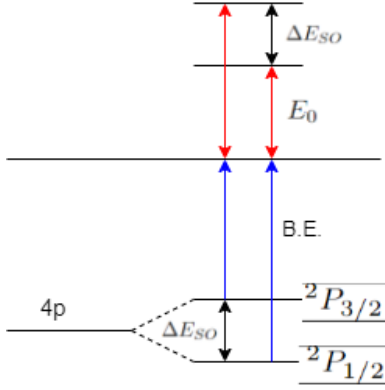


Fig. 7. Energy Diagram of the corresponding 4p sublevel. The red arrows are the Kinetic energies(E_0) and the blue ones represent the Binding Energies(B.E.). ΔE_{SO} is the fine structure splitting.

However, the booklet provided by Thompson [8] on the binding energies(B.E.) of atom's electrons, points that the N2 4p1/2 and N3 4p3/2 electrons have the same B.E. of 14.1 eV. These contradicts our scheme, where the BE is the same and thus the ΔE_{SO} cannot be observed since only the same K values are intercepted by their detector.

We don't know the detection capabilities and setup utilised in these measurements, but a further bibliographic search in the references send us to [2], from which these values are taken. These values are relevant for X-ray spectroscopy, where the wavelengths are in the range of 0.01 to 10 nanometers and energies in the range of 100 eV to 100 keV. Considering that these measurements were not made using HHG, it would explain that our measurements are a step forward in making more precise measurements with a different technique, and uncovering that these BE's have take into account the S.O. coupling.

The works of Stefan Pabs *et al.* [7], points to a $\Delta E_{SO} = 0.67\text{eV}$. If we were to compare our result to his we have to choose between averaging our results or choosing a result with smallest Resolution value. We choose the later. As we need to plot the Resolutions for comparison without program inherent error bars, we modify our python script by cancelling the error bars. Looking into figure 8 we can choose that the most stable as well as the smallest resolution trough-out the experiment was the 4th PES data acquisition. This makes sense when we compare the gaussian peaks in figure 4, where the most acute peaks also belongs to 4th data set.

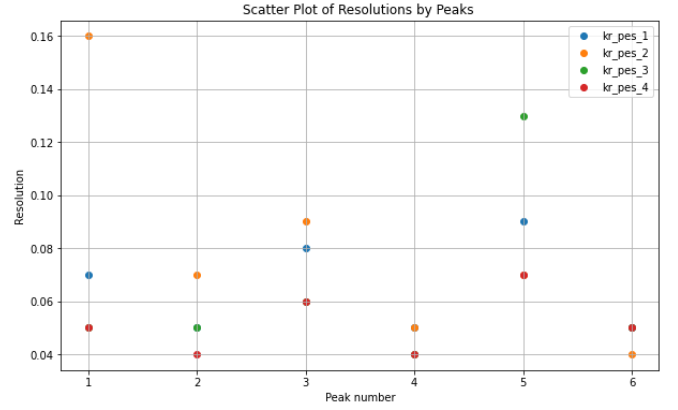


Fig. 8. Resolutions by peaks and data set acquisition

We have computed the Kr4Pes data set's S.O. splitting energies, and shown them bellow.

Peak	S.O. Splitting (eV)
1-2	0.57
3-4	0.46
5-6	0.52

TABLE IV
KR PES 4 S.O. SPLITTING ENERGIES

Although our values differ at least by 0.1 eV from the one by Stefan Pabst, his values are computer simulated with Time-Dependent Complete Active Space Self-Consistent Field methods. So a room for error or something else is afoot.

CONCLUSION AND PERSPECTIVES

In conclusion, our study of the electronic intricacies of Krypton gas using Photoelectron Spectroscopy (PES) and High-Harmonic Generation (HHG) has provided valuable insights into the spin-orbit splitting of Krypton electrons. The application of ultrafast techniques and precision measurements has allowed us to delve into attosecond dynamics, revealing the inherent complexities of photoionization processes.

The Gaussian fitting of the experimental data has facilitated the extraction of kinetic energy peaks, and our results show spin-orbit splitting energies of 0.57 eV, 0.46 eV, and 0.52 eV for the 1-2, 3-4, and 5-6 peaks, respectively. The resolutions of the spectrometer were also presented, shedding light on the precision of our measurements.

Our work positions itself within the state-of-the-art by recovering kinetic energy peaks and comparing them to precedent literature. By advancing the understanding of spin-orbit coupling in Krypton, we contribute to the broader field of quantum chemistry and spectroscopy. The utilization of HHG in our experimental setup represents a significant step forward, allowing for attosecond time domain exploration.

Despite the progress made, certain aspects remain unexplored. The comparison of our results with previous literature highlights discrepancies that warrant further investigation. Additionally, the complexity of the system and the limited time frame of the experiment have introduced uncertainties,

emphasizing the need for more extensive studies and improved measurement techniques.

Looking ahead, future research could focus on refining experimental setups, reducing uncertainties, and validating results through complementary methods. The discrepancies observed in binding energies between our results and literature suggest potential variations in detection capabilities and necessitate a deeper understanding of the interplay between experimental techniques and theoretical simulations of atomic and molecular dynamics.

ANNEX

Peak	A	E_0 [eV]	Std_err_to_Mean	σ	FWHM	Resolution
1	9834.40	16.51	0.0174	0.49	1.14	0.07
2	10645.77	17.18	0.0145	0.40	0.93	0.05
3	15391.52	19.92	0.0810	0.70	1.64	0.08
4	18422.65	20.31	0.0195	0.40	0.95	0.05
5	16721.35	23.28	0.2002	0.88	2.08	0.09
6	16594.20	23.39	0.0279	0.47	1.11	0.05

TABLE V
Kr PES 1

Peak	A	E_0 [eV]	Std_err_to_Mean	σ	FWHM	Resolution
1	9887.85	16.99	0.4973	1.18	2.78	0.16
2	9955.11	17.19	0.0309	0.52	1.23	0.07
3	15494.68	19.97	0.1676	0.78	1.84	0.09
4	18468.20	20.35	0.0188	0.39	0.92	0.05
5	15071.33	22.98	0.0836	0.65	1.53	0.07
6	17189.09	23.42	0.0310	0.44	1.05	0.04

TABLE VI
Kr PES 2

Peak	A	E_0 [eV]	Std_err_to_Mean	σ	FWHM	Resolution
1	6254.84	16.68	0.0382	0.36	0.84	0.05
2	7592.78	17.23	0.0290	0.37	0.86	0.05
3	14849.80	19.82	0.0215	0.50	1.18	0.06
4	18598.59	20.37	0.0163	0.37	0.87	0.04
5	16412.07	23.75	0.7866	1.29	3.04	0.13
6	14510.88	23.44	0.0380	0.54	1.26	0.05

TABLE VII
Kr PES 3

Peak	A	E_0 [eV]	Std_err_to_Mean	σ	FWHM	Resolution
1	19554.04	16.60	0.0079	0.32	0.75	0.05
2	24786.81	17.18	0.0141	0.29	0.68	0.04
3	51447.42	19.83	0.0291	0.52	1.22	0.06
4	66606.89	20.33	0.0143	0.36	0.85	0.04
5	52703.98	23.05	0.0858	0.71	1.67	0.07
6	58481.97	23.41	0.0285	0.46	1.09	0.05

TABLE VIII
Kr PES 4

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